

LUKE MAUZY

Electrical Engineering

EDUCATION

University of Virginia, Charlottesville, VA**Bachelor of Science in Electrical Engineering***Expected May 2028*

- ◆ GPA: 3.75 | Computer Systems and Organization, Signals and Systems, Electronics, Wireless for the Internet of Things, Circuits, Probability, Machine Learning

Maggie L. Walker Governor's School, Richmond, VA**High School Diploma***May 2024*

- ◆ Member of four academic honor societies (NHS, Spanish, German, Math); captain of varsity basketball and volleyball; 200+ hours of community service.

TECHNICAL SKILLS

- ◆ **Software:** Python, C, C++, MATLAB, Git/GitHub, KiCad, Fusion360, Powershell, VS Code, Microsoft Excel
- ◆ **Hardware:** Circuit Design, PCB Design, Soldering, 3D Printing, Arduino, Raspberry Pi, Oscilloscopes/Multimeters

EMPLOYMENT HISTORY

AUTOMATION SYSTEMS INTERN*Jun 2026 - Present***Siemens***Richmond, VA*

- ◆ Support Smart Industries Building Automation projects, gaining hands-on experience in HVAC controls configuration, programming, system commissioning, and troubleshooting.
- ◆ Perform end-to-end testing to validate system performance and resolve deployment issues across building automation systems.
- ◆ Leverage Microsoft Copilot as part of a companywide initiative to consolidate information across SharePoint and internal databases, supporting strategic planning and forward-looking project initiatives.
- ◆ Collaborate with cross-functional teams and building automation specialists to support the installation, troubleshooting, and servicing of complex control systems for commercial and industrial clients.
- ◆ Program and configure Siemens PXC and DXR controllers using platforms including ABT Site and DesigoCC, working across BACnet, Modbus, and proprietary communication protocols.
- ◆ Gaining a comprehensive understanding of the full project workflow, from initial sales and detailed engineering design through installation, commissioning, and ongoing service.
- ◆ Apply analytical and programming skills to diagnose and resolve control system malfunctions, contributing to efficient and reliable building operations.

UNDERGRADUATE TEACHING ASSISTANT — APPLIED CIRCUITS*Jan 2026 - May 2026***University of Virginia***Charlottesville, VA*

- ◆ Graded labs, homework, and exams for an Applied Circuits course, ensuring accurate, consistent, and timely evaluation of student work.
- ◆ Assisted students during hands-on lab sessions, troubleshooting circuit implementations and reinforcing core concepts in real time.

SUMMER INTERN*Jun 2025 - Aug 2025***Chewning and Wilmer Electrical Contracting Inc.***Richmond, VA*

- ◆ Assembled and wired large-scale battery cabinet systems providing redundant backup power for an AI data center, integrating high-capacity storage units to sustain operations during AC power loss.
- ◆ Applied electrical engineering principles in cabling, circuit connections, and system grounding; conducted continuity and functionality testing to meet design specifications and ensure safety compliance.
- ◆ Evaluated energy consumption patterns to optimize power distribution, leading to measurable improvements in system efficiency.

STUDENT RESEARCHER*Jan 2025 - May 2025***US Navy and UVA School of Engineering***Charlottesville, VA*

- ◆ Built MATLAB and Python-based simulations to optimize stage mass ratios for a three-stage, 250 kg solid-engine rocket, benchmarking total change in velocity per stage against a fixed 10s first stage burn control; co-authored a 50-page technical report on the methodology and findings.
- ◆ Applied the Tsiolkovsky equation to model and compare total velocity and range of 'popup' boosters against current Gas Management Systems in U.S. Navy shipboard missile launchers, showing short-duration boosters result in only marginal efficiency losses.

LEADERSHIP & ACTIVITIES

UVA CLUB VOLLEYBALL**Charlottesville, VA**

- ◆ Selected by peers as team captain; travel to other universities and cities for tournaments and exhibition games.

UVA SOLAR CAR TEAM**Charlottesville, VA**

- ◆ Member of a multidisciplinary team that engineered a solar-powered vehicle from scratch to compete in a nationwide collegiate solar car competition.
- ◆ Collaborated on the power sub team to design and assemble a battery pack that met strict competition safety and performance constraints.
- ◆ Conducted rigorous battery testing and analysis to optimize cell arrangement, improving power output and overall energy efficiency.
- ◆ Designed a structural mount using CAD to securely hold the battery box in place during high-acceleration conditions.
- ◆ Worked cross-functionally with mechanical and electrical sub-teams to integrate the power system into the vehicle's overall design.